

ACM Chapter SOA

Inside The Mind of a Hacker

Computer Science Department

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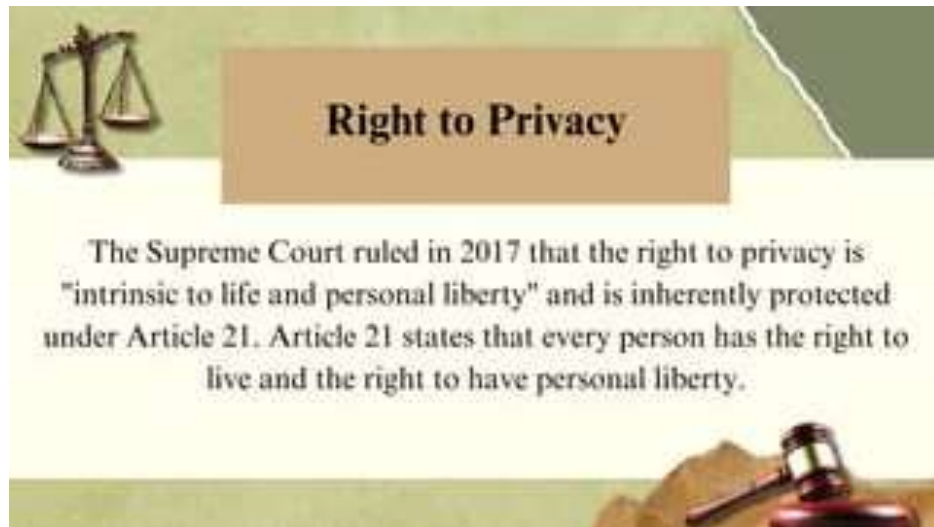
Dr. Rourab Paul

Cyber Security

Why This Matters

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- Your data = money
- Students are easy targets
- Cyber attacks happen every second



Quick Question

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- How many of you reuse passwords?
- How many use public WiFi?
- You are already a target.

Clicked “**Forgot Password**”

Answered simple questions:

- Date of birth
- High school name

Found answers via:

- Google
- Public information
(Wikipedia)



Sarah
Palin

What is Cybersecurity?

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- Protecting systems, data, and identity from attacks.
- Not just companies—YOU.

How Hackers Think

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- Look for weakest link
- Exploit human mistakes
- Automate attacks

Who is a Hacker?

- Not always criminal
- Problem solver
- Types:
 - White hat
 - Black hat
 - Grey hat

Common Attack #1: Phishing

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- Fake emails/login pages
 - Goal: steal your credentials
 - skills: website/app design
1. Download html, css, js of the login page of popular websites
 1. Register a domain name like faceb00k.com, facebook.com or anything. We hardly notice address bar.
 2. Upload the webpage in new domain. Share the link with targets
 3. Target may log in with tier credential, collect data, redirect to original website



Common Attack #2: Password Attacks

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- Weak passwords
- Brute force & dictionary attacks



1. Your Friends email: rahul123@gmail.com
2. checks: Instagram bio Birthday posts Pet names Favorite team : Example guesses: Rahul@123 Rahul2002 Messi@10
3. OSINTT tool to gather information
4. Automation Tool Software tries thousands of passwords per second
5. Automation may be prevented by Captcha

Common Attack #3: Social Engineering

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- Manipulating people
 - “Hi, I’m from your bank...”
1. Information Gathering – collect victim details (name, phone, bank, social media)
 1. Impersonation – pretend to be a trusted entity (bank, support, friend)
 1. Trust Building – use convincing language and context
 2. Trigger Action – create urgency (OTP, password, link click)
 3. Access Gained – victim shares sensitive information
- Exploitation – misuse data (fraud, account takeover)



Live Thinking

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- Would you click this link?
- Would you trust this email?
- Think before you act.

Real Impact

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- Money loss
- Identity theft
- Reputation damage

Ukraine power grid cyberattack 2015

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In 2015, hackers attacked Ukraine's power system

Around 230,000 people lost electricity

How the attack worked

1. Phishing emails

- Employees received infected emails

2. Malware installed

- Hackers gained access to systems

3. Remote control

- Took control of power grid computers

4. Power shutdown

- Turned off electricity remotely

WannaCry ransomware attack 2016

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Spread across 150+ countries

Locked computers and demanded money

How the attack worked

- Exploited **unpatched Windows systems**
- Spread automatically (worm)

Hospitals, banks, companies affected

UK healthcare system disrupted

Tools Hackers Use Generally

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- Automation tools
- Password crackers
- Phishing kits

Cybersecurity Careers

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- Ethical Hacker
- Security Analyst
- Bug Bounty Hunter

Cybersecurity Careers

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- **Computer fundamentals** (how OS, files, and programs work)
- **Networking basics** (IP, DNS, HTTP/HTTPS)
- **Operating Systems** (especially Linux basics)
- **Programming basics** (Python / C / scripting)
- **Web fundamentals** (how websites and logins work)
- **Cybersecurity basics** (common attacks like phishing, malware)

Cybersecurity Careers

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- **Nmap – network scanning & port discovery**
- **Wireshark – analyze network traffic**
- **Burp Suite – test web applications**
- **Metasploit – exploit vulnerabilities**
- **John the Ripper – password testing**
- **Aircrack-ng – WiFi security analysis**

Cybersecurity Certifications

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□ Beginner Friendly

- **Google Cybersecurity Professional Certificate**
→ Good starting point, no prior experience needed
- **Introduction to Cyber Security**
→ Basics of threats, networks, security

● Hands-on Learning (Highly Recommended)

- **TryHackMe**
→ Interactive labs, beginner to advanced

- **Hack The Box**

Cyber Security

□ Certification (for career growth)

- **CEH (Certified Ethical Hacker)**
→ Popular but more theory-based
- **CompTIA Security+**
→ Strong foundation, industry recognized

● Advanced (later stage)

- **OSCP (Offensive Security Certified Professional)**
→ Highly respected, hands-on, tough

OWASP

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Open Worldwide Application Security Project (OWASP)

- A global **non-profit organization** for web security
- Provides **top security risks & best practices**
- Used by companies worldwide

In simple words:

“OWASP tells us what mistakes make websites hackable.”

OSINTT

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OSINT (Open Source Intelligence)

- Collecting information from **public sources**
- No hacking required
- Uses:
 - Google
 - Social media
 - Public records

In simple words:

“OSINT = finding secrets that people accidentally make public.”

Post Quantum Cryptography

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https protocol applies cryptography

The screenshot shows the ING Belgium website. The browser's address bar displays the URL `https://www.ing.be/en/retail/Login`, with the `https` and the domain `www.ing.be` circled in red. The page features the ING logo and a navigation menu with options like 'Individuals', 'Business', and 'Other sites'. Below the navigation bar, there's a section titled 'Login to Home'Bank' with a form for user authentication. The form is divided into two sections: '1. Your details' and '2. Identification'. The '1. Your details' section includes fields for 'ING ID', 'Card ID', and 'Password', along with radio buttons for 'Home'Bank' and 'Business'Bank', and a 'Save my details' checkbox. The '2. Identification' section includes instructions for using an ING bank card and a PIN, with a visual representation of the card and a PIN entry field.

ING BELGIQUE SA [BE] `https://www.ing.be/en/retail/Login`

Individuals Business Other sites ▾ OUR BRANCHES | CONTACT

ING

EN | FR | NL

Day-to-day banking Lending Savings Investments Insurance Retirement

Home > Login to Home'Bank

Login to Home'Bank

1. Your details

ING ID ⓘ Enter Your ING ID

Card ID ⓘ 6703

☒ Home'Bank ☐ Business'Bank

☐ Save my details

Password ⓘ

[New password](#) | [Forgot your password?](#)

2. Identification

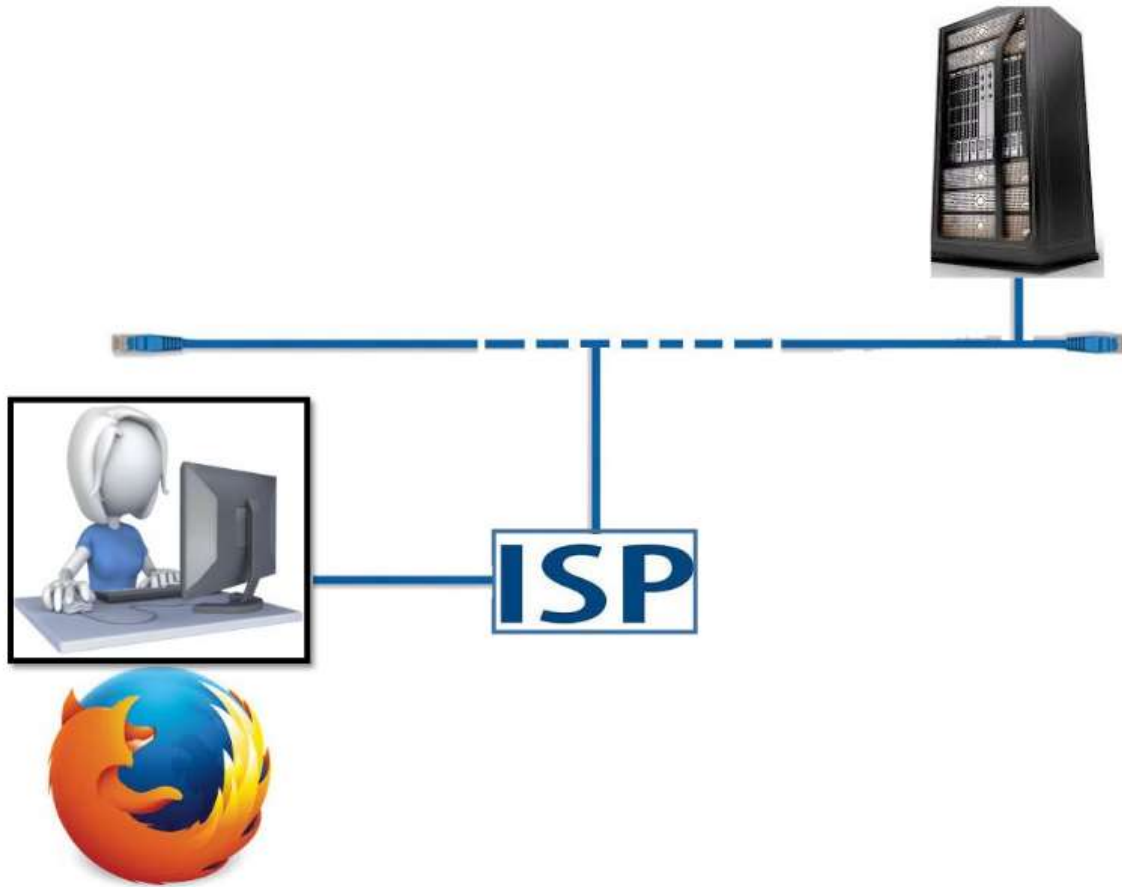
1. Insert your ING bank card into the ING Card Reader and press: **IDENTIFY**

2. Enter the pin of your ING bank card and press: **OK**

3. Enter the number appearing on the ING Card Reader screen, without spaces.

Post Quantum Cryptography

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Post Quantum Cryptography

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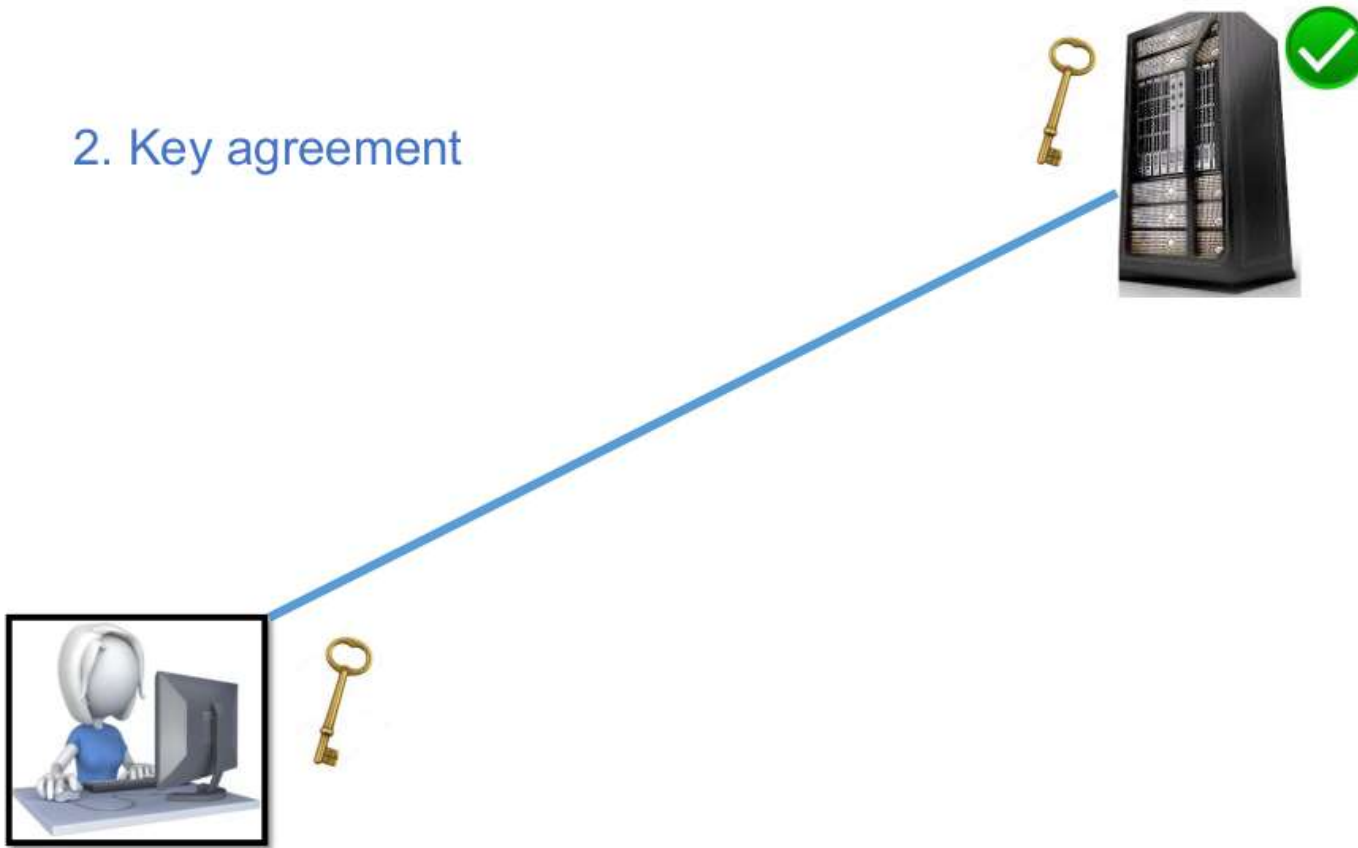
1. Identification of the website



Post Quantum Cryptography

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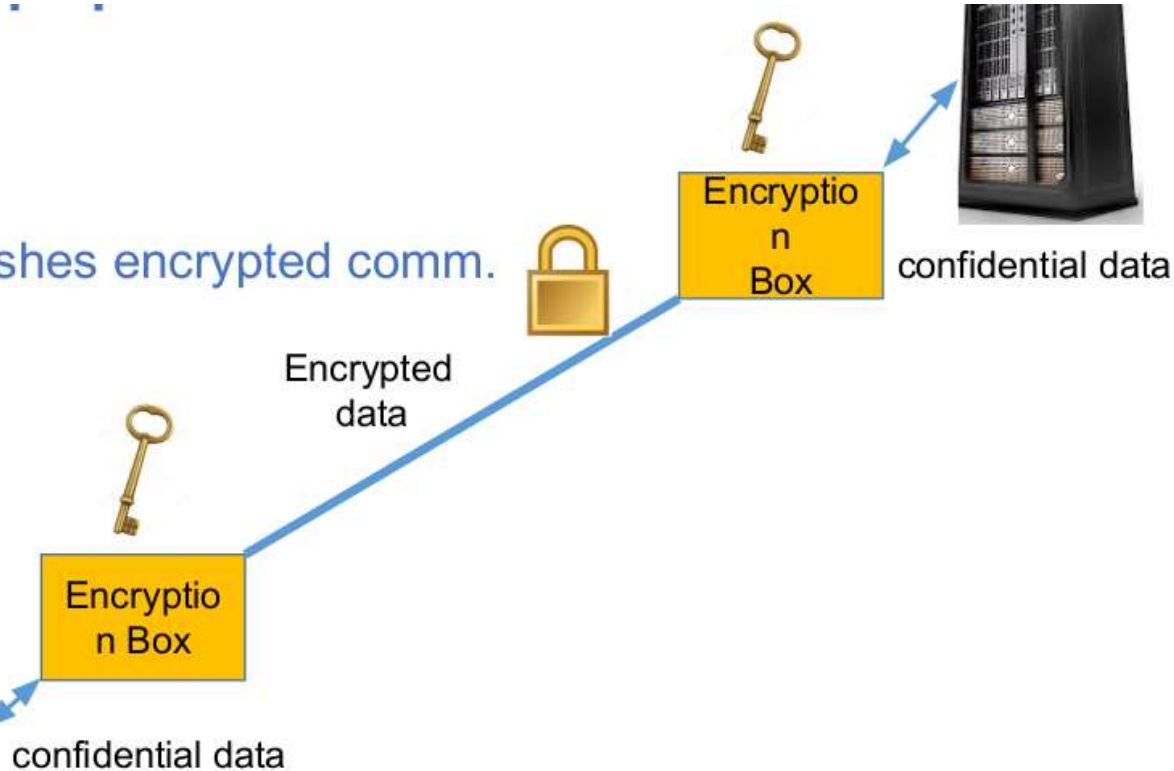
2. Key agreement



Post Quantum Cryptography

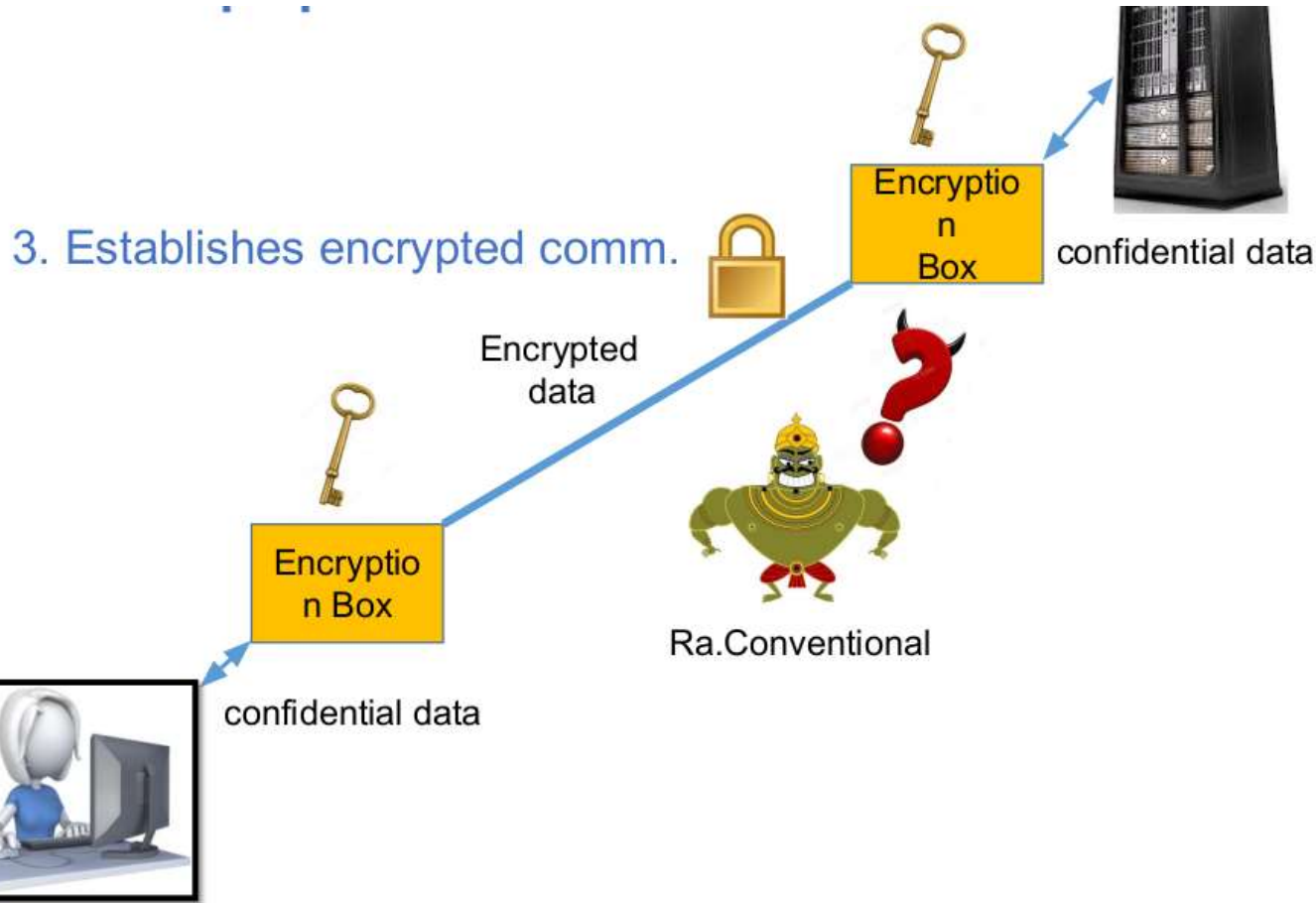
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3. Establishes encrypted comm.



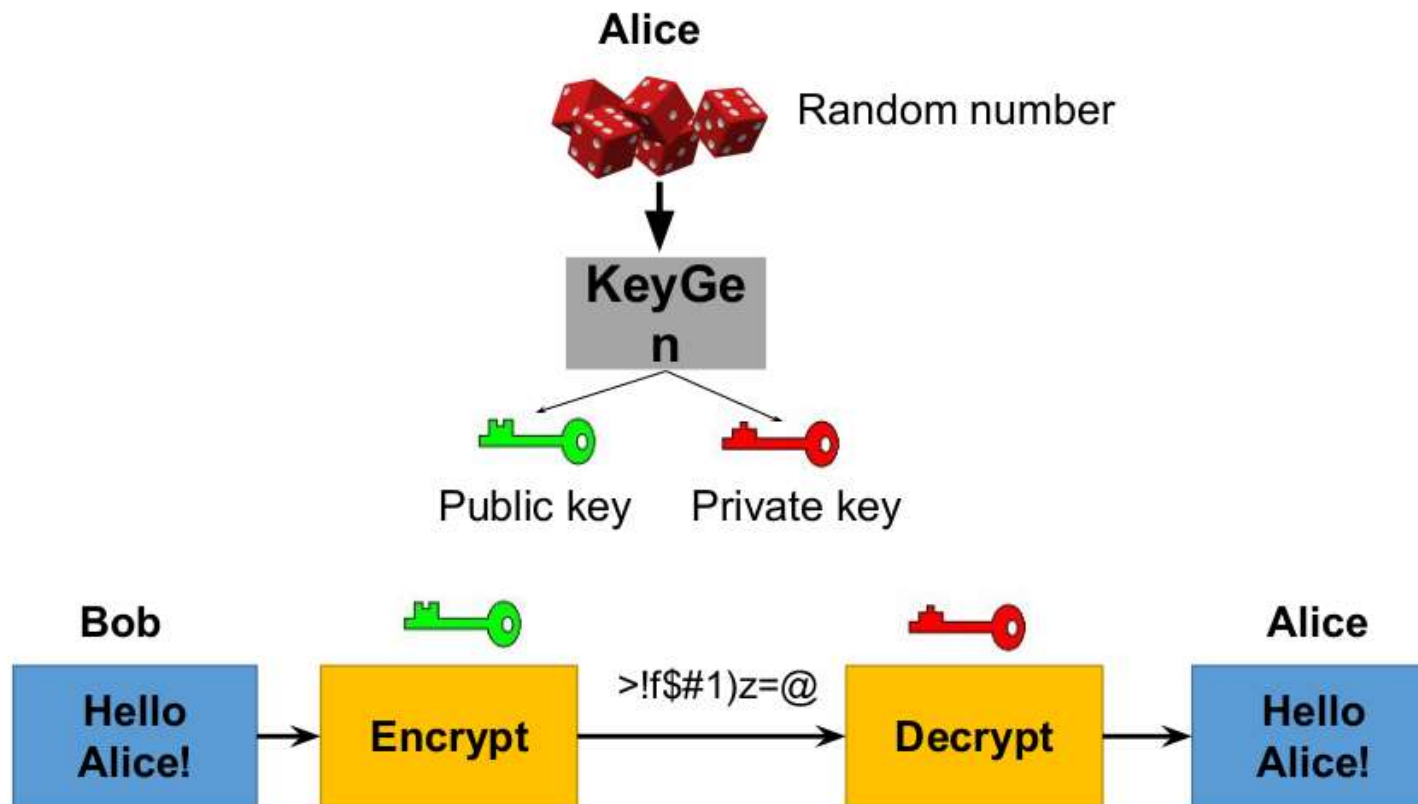
Post Quantum Cryptography

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Public Key Cryptography

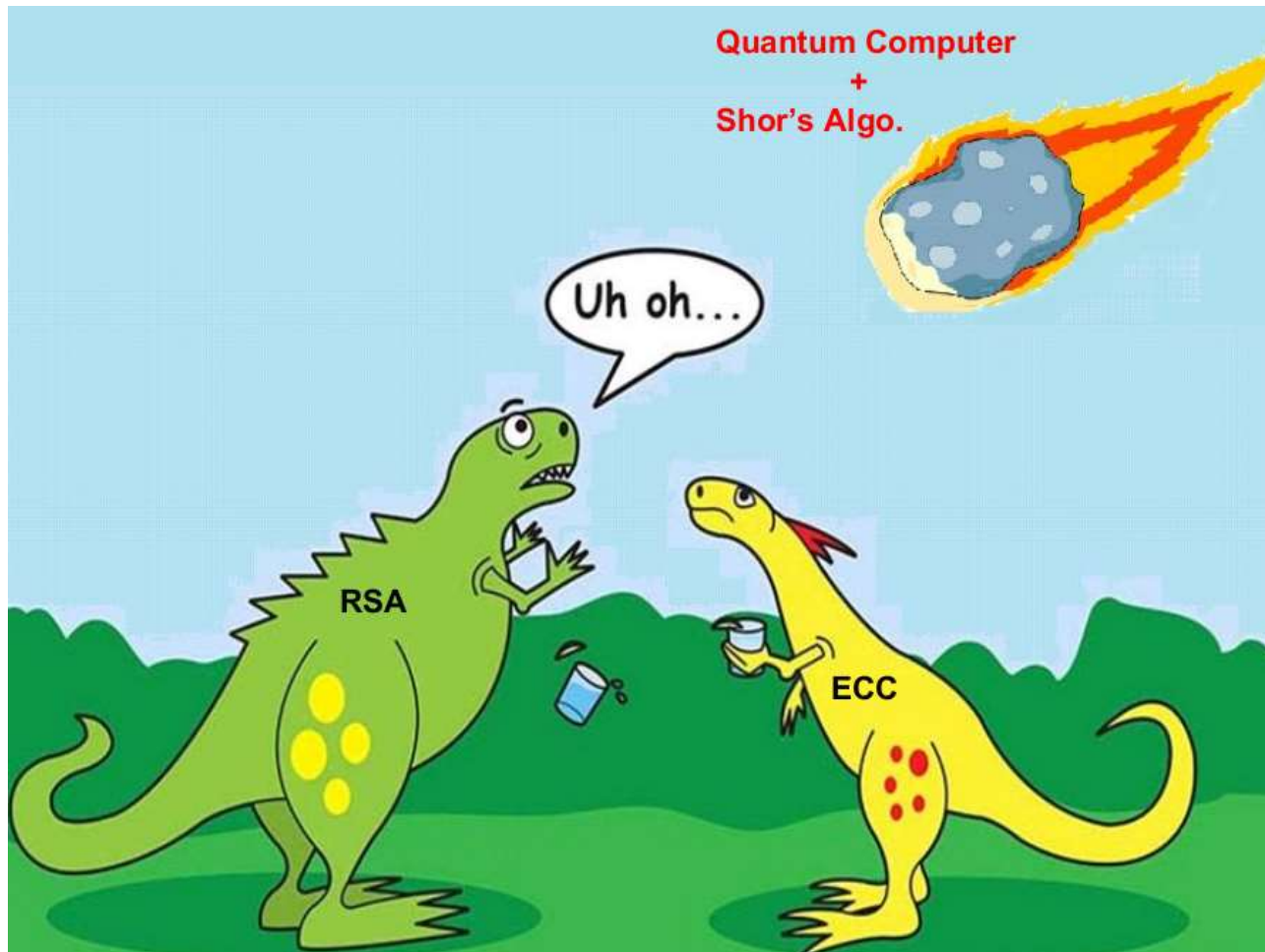
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Public key encryption

Public Key Cryptography

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Cyber Security

Hackers don't hack systems...

They hack humans.

Be aware. Stay secure.

Thank You